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Time to Practice: Non-Negative Matrix Factorization (NMF)
Graded Assignment • 15 min

Due Oct 26, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

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1. In the context of using NMF for topic modeling, what does the constraint that the matrices W and H must be non-negative imply about the topics? 1 / 1 point

- Topics can include negative feature weights for differentiation
- Topics are independent and do not share common features
- Topics are represented as additive combinations of non-negative features
- Topics are represented as subtractive combinations of features

✓ **Nice work**

The non-negativity constraint in NMF ensures that the topics and their corresponding term weights are represented as additive combinations of features, which can provide a more interpretable parts-based representation of the text content.

2. In the context of using NMF for topic modeling, how would you identify the most relevant terms for a specific topic? 1 / 1 point

- Take the smallest values in the corresponding row of matrix H.
- Multiply the W and H matrices and for each row, select the terms with highest weight.
- Sort the values in the corresponding row of matrix H and select the top terms.

✓ **Nice work**

The H matrix in NMF represents the term weights for each topic. By sorting the values in a specific row, which corresponds to a topic, and selecting the top terms, you can identify the most relevant terms for that topic.

3. What does the H matrix represent in NMF topic modeling? 1 / 1 point

- Document-topic weights
- Term frequency matrix
- Document similarity scores
- Topic-term weights

✓ **Nice work**

In NMF, the matrix H contains the weights of terms for each topic, where each row corresponds to a topic and each column corresponds to a term in the corpus vocabulary.

4. What is the purpose of the W matrix in NMF? 1 / 1 point

- To calculate the error between the original and reconstructed matrices
- To represent topic scores for each document
- To normalize the document vectors in the dataset
- To store the original data before factorization

✓ **Nice work**

The W matrix in NMF has one row per document, with each column representing the topic score for that document, indicating how relevant each topic is to the document.

5. How can you find the most representative documents for a topic using NMF? 1 / 1 point

- Multiply the corresponding column of matrix W by the transpose of matrix H
- Calculate the average of all entries in the corresponding row of matrix H.
- Sum the values in the corresponding column of matrix W and select documents based on this sum.
- Sort the values in the corresponding column of matrix W and select the top documents.

✓ **Nice work**

The W matrix in NMF contains the topic scores for each document. Sorting the values in a column, which corresponds to a topic, and selecting the top documents will show which documents are most representative of that topic.